MUSTHOFA JOKO ANGGORO

musthofaja.topa@gmail.com | linkedin.com/in/musthofa-joko-anggoro | github.com/topahilangharapan | portfolio-topahilangharapan-personal.vercel.app

RESEARCH INTERESTS

My research centers on **reconfigurable computing and FPGA-based hardware acceleration**, with particular interest in mapping compute-intensive workloads onto resource-constrained programmable devices. Through my undergraduate thesis — a full RTL implementation of the Mano register-transfer model on a Xilinx Spartan-3AN FPGA in VHDL-93 — I developed hands-on expertise in datapath design, RTL synthesis, place-and-route closure, and hardware verification. I am drawn to research at the intersection of **FPGA architectures, edge AI hardware, and reconfigurable systems**, asking how hardware-aware design methodologies and accelerator architectures can enable efficient, low-power inference at the edge. I am enthusiastic about contributing to the Hardware & Embedded Systems Lab (HESL) under Assoc Prof Lam Siew Kei and advancing the lab's work on FPGA-based accelerators and reconfigurable edge platforms.

EDUCATION

University of Indonesia (UI) — Faculty of Computer Science (Fasilkom) Jakarta, Indonesia
Bachelor of Computer Science, Information Systems *Aug 2022 – Aug 2026 (Expected)*

- **GPA: 3.84 / 4.00**
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS Singapore & Surabaya
INSPIRASI–NTU Summer Programme 2025 — **LPDP Scholarship Awardee** *July 2025*

- Applied software engineering and systems principles to sustainable development challenges; produced a technical report proposing a tiered smart-port automation framework integrating IoT sensor networks and renewable energy utilization

UNDERGRADUATE THESIS

VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on FPGA

Supervisor: Prof. Petrus Mursanto, University of Indonesia *Status: Completed — Paper in preparation for ICACSIS 2026*

- Designed and implemented the Mano register-transfer computer model as a complete, hierarchically structured VHDL-93 (IEEE 1076-1993) datapath on a **Xilinx Spartan-3AN** FPGA
- Conducted a systematic **gap analysis** mapping 5 undefined textbook constraints to concrete RTL design decisions, bridging theory and physical hardware implementation
- Implemented a **three-tier component hierarchy**: leaf-level primitives (`reg_n`, ALU, shifter) → functional units → full system integration (`fpga_top`)
- Verified all modules via **ISim functional simulation** testbenches and board-level testing on physical hardware
- Achieved **timing closure at 50 MHz** on the Spartan-3AN device (critical path: 17.05 ns) using XST synthesis engine
- Demonstrated live operation via PS/2 keyboard input and **HD44780 LCD display** output; programmed via JTAG
- *Toolchain:* Xilinx ISE 14.x · XST · ISim · JTAG · VHDL-93

PUBLICATIONS

- Musthofa Joko Anggoro, Petrus Mursanto. "VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on an FPGA." *Proceedings of the International Conference on Advanced Computer Science and Information Systems (ICACSIS) 2026.* [**In preparation**]

TECHNICAL SKILLS

HDL & Digital Design: VHDL-93 (primary), RTL synthesis, place-and-route, functional simulation, testbench writing, hierarchical hardware decomposition
FPGA Tools: Xilinx ISE 14.x · XST synthesis engine · ISim · JTAG; **Target Device:** Xilinx Spartan-3AN (XC3S700AN)
Systems Programming: Python, Java, C, Go, TypeScript
Machine Learning: Scikit-learn, supervised/unsupervised learning, model evaluation
Infrastructure: Docker, Kubernetes, CI/CD pipelines, Linux/Unix, Git